
Exercise Sheet 3:
Constraint Satisfaction Problems

Foundations of Neuro-Symbolic AI
Summer term 2026, THWS, Alex Goessmann

May 11, 2026

n queens problem

For arbitrary $n \in \mathbb{N}$ let there be a grid of $n \times n$ positions with n columns and n rows. We want to assign r queens onto positions of the chess board, such that:

- Each column contains exactly one queen.
- Each row contains at most one queen.

Task 1:

Design the n queens problem as a Constraint Satisfaction Problem.

- (i) What are suitable variables, their leg dimensions, and the constraint tensors?
- (ii) Which hidden variables could be introduced for a sparser representation of the constraints?

Hint: Compare the problem to the Sudoku game.

Task 2:

Assume again that we have a partial assignment of $r \leq n$ queens given. How many assignments of the remaining $n - r$ queens satisfy the constraints? How to compute that number by a tensor network contraction?

Task 3 (Generalized unit propagation):

Assume that we have a partial assignment of $r < n$ queens given, such that each pair of queens is assigned to different columns and different rows. Would the generalized unit propagation algorithm be able to find a solution for the remaining $r - k$ queens?

Task 4 (Backtracking):

Assume that we have a partial assignment of $r < n$ queens given, such that each pair of queens is assigned to different columns and different rows. How would the backtracking algorithm with generalized unit propagation as a simplifier find a solution for the remaining $r - k$ queens?